

Mat().createAIJ()

Excerpt from code,

```
I = PETSc.Mat().createAIJ([N*N, N*N])
I.setUp()
I.assemble()
```

The `.createAIJ` function is a wrapper for the `MatCreateAIJ()/MatCreateSeqAIJ()` function. The function is,

```
MatCreateSeqAIJ(MPIComm comm, PetscInt m, PetscInt n, PetscInt nz, const
                PetscInt nnz[], Mat *A)
```

Where each of the input and output parameters are,

- `comm` - MPI communicator
- `m` - number of rows
- `n` - number of columns
- `nz` - number of non-zero rows
- `nnz[]` - array containing the number of non-zero rows
- `A` - the matrix outputted

This function is used to create a matrix.

Sources from GitHub/Documentations;

- [petsc/petsc/-/blob/main/doc/manual/ksp.md](https://petsc.org/main/doc/manual/ksp.md)
- [petsc/petsc/-/blob/main/src/ksp/ksp/interface/itcreate.c](https://petsc.org/main/src/ksp/ksp/interface/itcreate.c)